

Polarization Path - Light Through Filters

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Abstract

This project focuses on creating and analyzing an AI Physics Lab simulation called **“Polarized Light - From Wave to Image”**, made by **Claude AI** running on **Sonnet 5**, that demonstrates the polarization and filtering of electromagnetic waves. The simulation uses a 3D interactive model of a projector, polarizing filters, and a screen to show how changing the filter angles affects the final light intensity and appearance. By rotating the filters, users can observe how waves are filtered and understand the relationship between filter angles and light intensity using concepts such as Malus's Law. The relationship is described by the formula:

$$I = I_0 \cos^2(\theta)$$

where I represents the transmitted light intensity, I_0 represents the initial light intensity, and θ represents the angle between the light polarization direction and the filter axis. Compared to some existing simulations that may lack 3D features or a clear final light display, this project combines interactive controls, visible wave movements, formulas, and intensity graphs to make polarization easier to understand. The model demonstrates values such as I/I_0 and shows how different filter positions change the amount of light passing through. Although more features, such as adjustable numbers of filters and additional electromagnetic wave properties, could be added, the current model provides a clear, interactive, and visually interesting way to learn about polarization and electromagnetic wave filtering.

Introduction

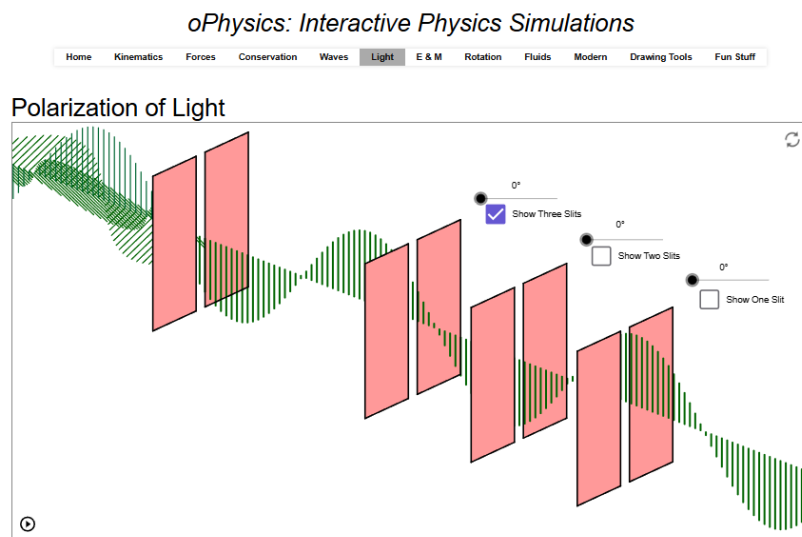


Figure 1: Snapshot of oPhysics' Polarization of Light simulation showing the visualization of polarized light passing through filters and the relationship between light orientation and polarizer axes.

This section discusses and reviews the details of the AI Physics Lab project. The physics concept behind this project is polarization and how electromagnetic waves from different sources can be filtered based on their axis of orientation. Other simulations, such as oPhysics' Polarization of Light simulation, lack the three-dimensional aspect and do not provide a final representation of the resulting light pattern on a viewing board (see Fig. 1; oPhysics). VisuPhy's 3D Light Polarization Simulation offers a more feature-rich environment, but the interface contains several non-essential visual elements, such as plane axes, component labels, and an extensive optical path table with setup and output parameters. While these features may be useful for advanced analysis, they are not essential for a basic

understanding of polarization and can make the simulation harder to interpret. The simulation also provides little explanation of polarization concepts, placing greater emphasis on configuring the simulation than on helping users understand the underlying principles. In addition, interaction within the 3D environment is not always smooth or intuitive, increasing the difficulty of exploring the simulation effectively (see Fig. 2; VisuPhy). It could be improved by making the interactions more intuitive and by including a visualization of the final light output on a board. Both simulations have areas where they can be improved.

The topic of electromagnetic waves can be challenging to understand due to their three-dimensional nature and because they involve the combination of electric fields, magnetic fields, and wave behavior. Therefore, simulations of electromagnetic waves should be smooth, understandable, and provide users with an appropriate level of control over the interactions.

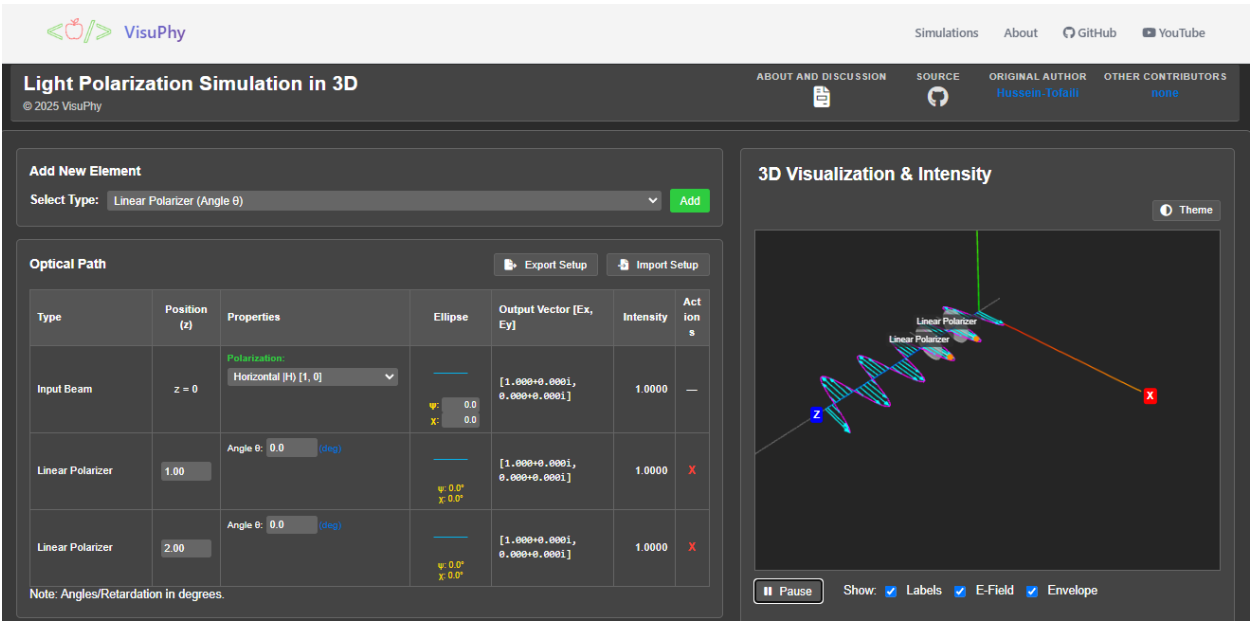


Figure 2: Snapshot of VisuPhy's Light Polarization Simulation in 3D showing the three-dimensional representation of polarized light waves and the interaction between the wave model and the simulation environment.

Methods

The models tested for this project were Claude running on Sonnet 5 and ChatGPT running on GPT-5.5. Although both models were used with approximately the same token limits, had not been used for several days beforehand, and were accessed through their standard free plans, their performance differed significantly. ChatGPT did not successfully follow the second prompt. Instead, it indicated that the response would be too lengthy to provide in a single attempt, requiring an alternative approach. To address this limitation, the project was divided into smaller sections that could later be combined. However, this method was also unsuccessful, as some of the generated code sections conflicted with previously provided sections. Additionally, attempts to revise and improve the project with the model were unsuccessful, even after at least four iterations.

In contrast, Claude generated the simulation directly from the provided prompt without requiring additional clarification. The resulting visual effects were significantly better than ChatGPT's initial attempts, including improved sizing, framing, and overall presentation quality.

The original two prompts were used with both models. These prompts were sufficient for Claude to produce the desired simulation; however, ChatGPT required significantly more interaction and adjustments without ultimately producing a successful result.

Prompt 1

"Write a simulation for polarization. There is a projector that shines a single electromagnetic wave, and then there is a filter that you can rotate by moving a bar. Just an HTML code with well-designed CSS and functional JavaScript."

Prompt 2

"Make it 3D. Make it more interactive by adding more waves (maybe 3 total) and at least one more filter."

Results

The final simulation developed for this project is a three-dimensional model consisting of a projector, a screen, and two polarizing filters positioned between them. This setup generates three electromagnetic waves that travel through the system, allowing users to observe how the rotation of each filter affects the amount of transmitted light. By adjusting the orientation of the filters, different portions of the waves are filtered, and the resulting light pattern can be observed on the screen. For example, rotating either filter by 90 degrees completely blocks the transmission of the waves, resulting in no light appearing on the screen.

The simulation provides an interactive 3D environment that is easy to navigate and allows the motion of the electromagnetic waves to be clearly observed. It also includes descriptions and relevant formulas to help guide users through the concepts being demonstrated. Additionally, the simulation contains a graph displaying the relationship between the final light intensity ratio I/I_0 and the filter angle θ_1 . The filters can be adjusted through both direct interaction with their 3D models and through the slider controls located next to the simulation. Figures 3–6 demonstrate how the relative orientation of two polarizing filters affects light transmission according to Malus' Law. When both filters are aligned at the same angle (0° in Figure 3 or 45° in Figure 5), their transmission axes are parallel, allowing the maximum possible light to pass after the initial reduction by the first filter, resulting in normalized intensities of 1.0 and 0.5, respectively. When the filters are oriented at different angles, the transmitted intensity decreases: a 45° difference (Figure 4) reduces the final intensity to 0.25, while a 90° difference (Figure 6) makes the transmission axes perpendicular, completely blocking the light and producing zero transmitted intensity. These results illustrate that the transmitted light depends on the relative angle between the polarizers rather than their absolute orientation.



Figure 3: Both polarizing filters are aligned at 0° , allowing the maximum amount of light to pass through the system. Since the transmission axes of both filters are parallel, the light intensity remains at 100% of the initial intensity ($I/I_0 = 1$), as shown by the graph.

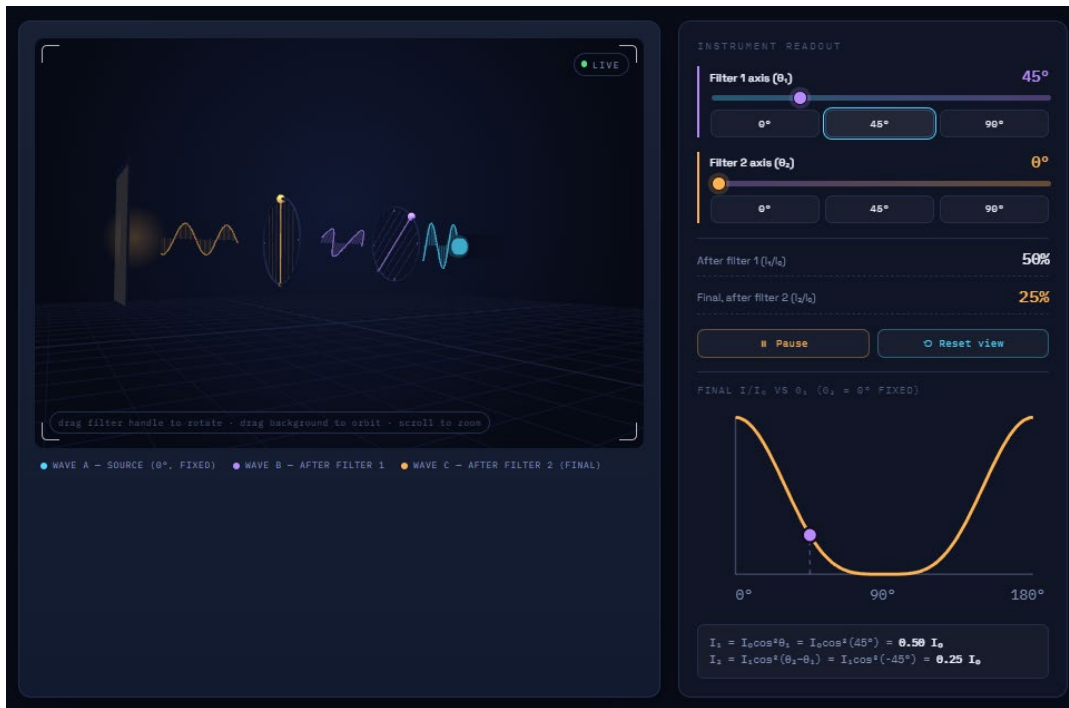


Figure 4: The first polarizing filter is rotated to 45° while the second filter remains at 0°. The angle difference between the filters reduces the transmitted light intensity according to Malus' Law, resulting in approximately 50% intensity after the first filter and 50% of that remaining light after the second filter, producing a final intensity of 25% ($I/I_0 = 0.25$).

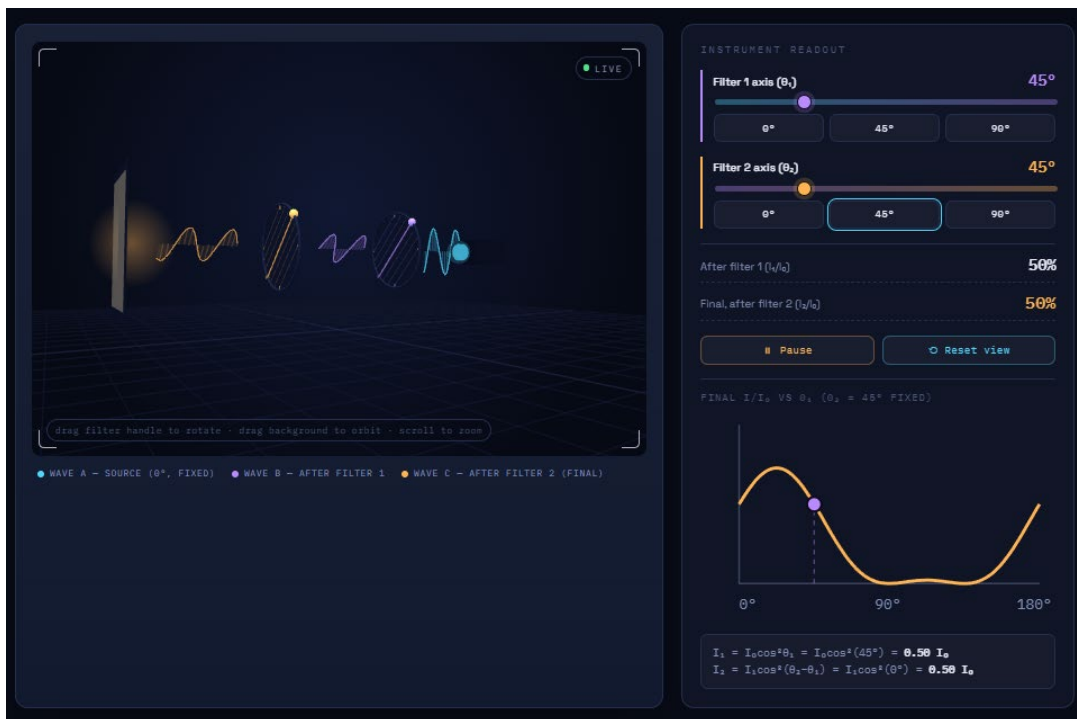


Figure 5: Both polarizing filters are rotated to 45°, maintaining parallel transmission axes. Because the relative angle between the filters is 0°, no additional filtering occurs between them; however, the first filter reduces the initial unpolarized light intensity by half, resulting in a final intensity of approximately 50% ($I/I_0 = 0.5$).

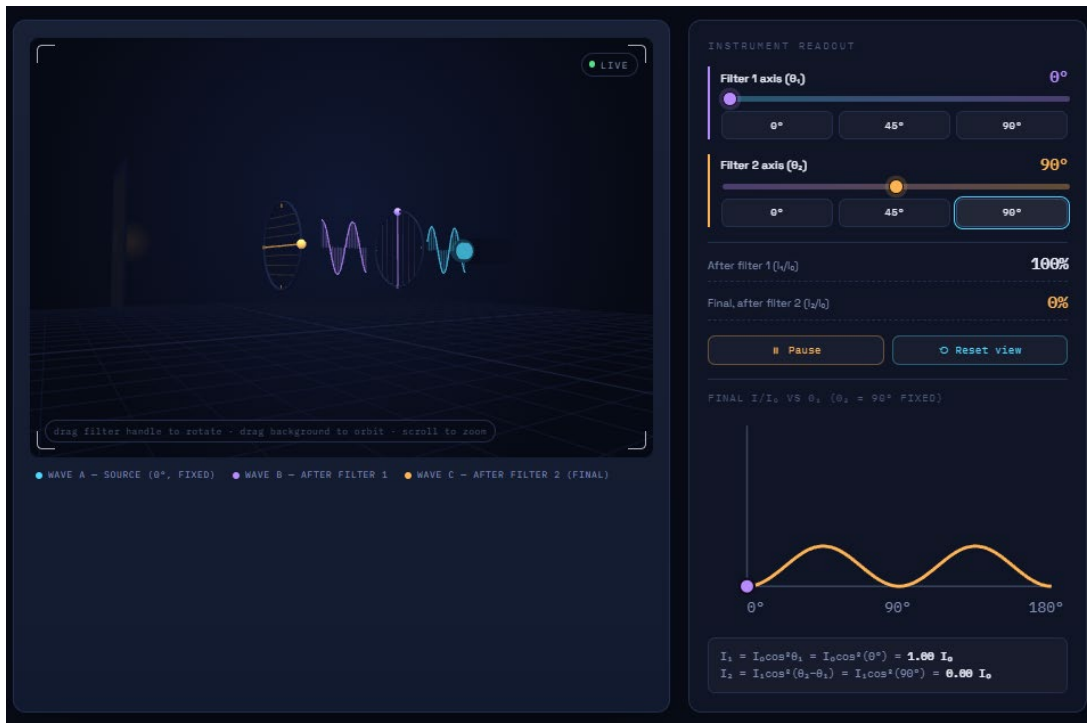


Figure 6: One of the polarizing filters is rotated to 90° relative to the other filter, causing the transmission axes to become perpendicular. This orientation completely blocks the polarized light from passing through the second filter, resulting in zero transmitted intensity ($I/I_0 = 0$). The same effect occurs if the second filter is rotated to 90° while the first filter remains unchanged.

Discussion

This model and simulation are sufficient for demonstrating the fundamental concepts of polarization and the filtering of electromagnetic waves. The simplified design allows users to clearly understand the relationship between filter orientation and the resulting light intensity. However, the number of polarizing filters in the simulation is currently fixed and cannot be adjusted by the user. While the limited number of features helps maintain an easier-to-understand representation, additional features related to electromagnetic wave behavior could be implemented to provide a more complete simulation experience.

The simulation also displays the resulting light intensity as a percentage by calculating the effects of both filters using the intensity relationships for filter 1: I_1/I_0 and filter 2: I_2/I_0 . This provides users with a clearer connection between the filter angles and the final transmitted light intensity. Future improvements could include adjustable numbers of filters, additional electromagnetic wave properties, and more advanced interactions while maintaining an intuitive and accessible design.

Conclusion

In general, this lab simulation provides an accessible way to understand and interact with the filtration of electromagnetic light waves. It allows users to observe how the orientation of polarizing filters affects the transmission of light and provides a visual connection between the theoretical equations and their physical behavior. Although many additional features could be implemented, the simulation could also be expanded into a larger model that represents electromagnetic waves through a wider range of characteristics, behaviors, and applications in different environments.

Despite its simplified design, the simulation remains easy to operate, visually engaging, and technically accurate in its representation of the underlying physics. The relationship between filter angles and transmitted light intensity is calculated throughout the simulation using the equation:

$$I = I_0 \cos^2(\theta)$$

where the initial unpolarized light intensity is reduced after passing through the first polarizing filter. Additionally, the equation:

$$I_2 = I_1 \cos^2(\theta_2 - \theta_1)$$

is used to determine the intensity after the second filter based on the relative angle between the two filters. These equations allow the simulation to accurately demonstrate the principles of polarization and provide a direct visualization of how changes in filter orientation influence the final observed light intensity.

References

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